

Unit 4, Lesson 1: Understanding Set-Builder Notation



Benchmark

MA.912.F.1.5 Compare key features of linear functions each represented algebraically, graphically, in tables or written descriptions.



Learning Targets

- ☐ I can understand the domain and range of a linear relationship.
- ☐ I can write inequalities using set-builder notation.



4.1.1 Warm-Up: What Is Infinity?

In *Toy Story*, Buzz Lightyear uses the phrase “To infinity and beyond.” But what does it mean to go beyond infinity? Infinity is an elusive concept that can often be difficult to comprehend. Eugenia Cheng, a mathematician, explains infinity as “not a normal kind of number.” Eugenia uses set theory to explain infinity. She uses the idea of a bag as a set. By placing a bag in a bag, there is a bag with one thing. Infinity can be illustrated by adding more and more bags and each bag contains other bags. Because you can keep adding bags with no end, the set contains an infinite number of bags.



1. Explain briefly your understanding of infinity.

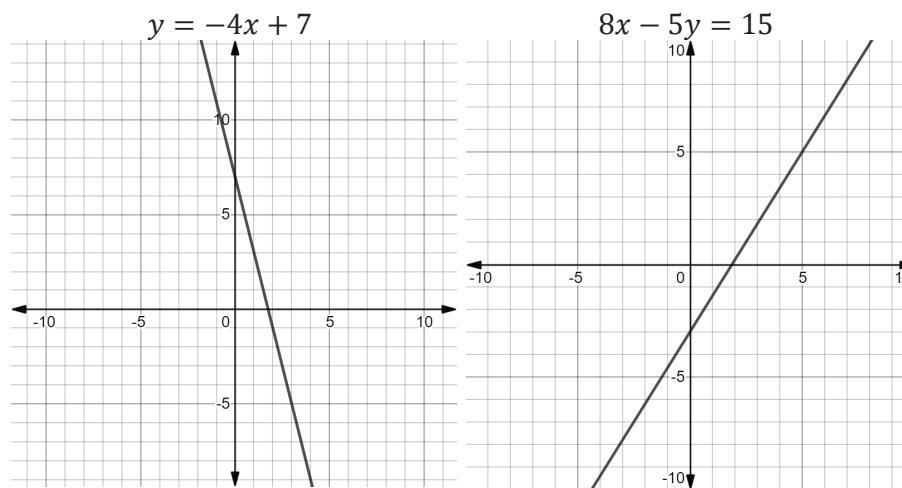
2. In the video, Eugenia Cheng uses the word paradox. A paradox is a seemingly absurd or self-contradictory statement that when investigated may prove to be well-founded, or true. Why do you think the idea of infinity can create a paradox?

3. The symbol for infinity is ∞ . The infinity symbol was first used by John Wallis in 1655. Draw an infinity symbol without picking your writing utensil up from the paper.
4. How many times can you trace over the infinity symbol without picking up your writing utensil from the paper?



4.1.2 Collaboration: Domain and Range of Linear Relationships

The graphs of $y = -4x + 7$ and $8x - 5y = 15$ are shown.



A **line** is a straight path that extends infinitely in both directions.

The graphs of $y = -4x + 7$ and $8x - 5y = 15$ are lines.

1. Two students were discussing how to determine if there is an x -value(s) not included in the graph of $y = -4x + 7$. Their explanations are shown.

Quanasia	Lysha
The graph of $y = -4x + 7$ is steep so it will never reach values such as $x = 10$. I think it eventually reaches $x = 5$, but I don't think it reaches many x -values after that.	The graph cannot show the entire line. If I input any value in for x , the output, y , will always be a number.

Discuss with your partner the two different explanations. Then, use the definition of a line to justify whether there is an x -value(s) not included in the graph of the line $y = -4x + 7$.



The **domain** is the complete set of possible values of the input of a function or relation. The domain may vary depending on the context.

2. Based on the definition of domain, what values can be used as an input for x if $y = -4x + 7$?
- (A) all values greater than -5 and less than 5
 - (B) all values greater than -20 and less than 20
 - (C) it cannot be determined
 - (D) any value

3. Based on the definition of domain, what values can be used as an input for x if $8x - 5y = 15$?
- (A) all values greater than -5 and less than 5
 - (B) all values greater than -20 and less than 20
 - (C) it cannot be determined
 - (D) any value

4. List any y -value(s) not included in the graph of $y = -4x + 7$. If none, then explain why.

5. List any y -value(s) not included in the graph of $8x - 5y = 15$. If none, then explain why.



The **range** is the complete set of possible values of the output of a relation or function.

6. Based on the definition of range, for the linear relationship $y = -4x + 7$, what values are outputs for y ?

- Ⓐ all values greater than -10 and less than 15
- Ⓑ all values greater than -20 and less than 20
- Ⓒ it cannot be determined
- Ⓓ any value

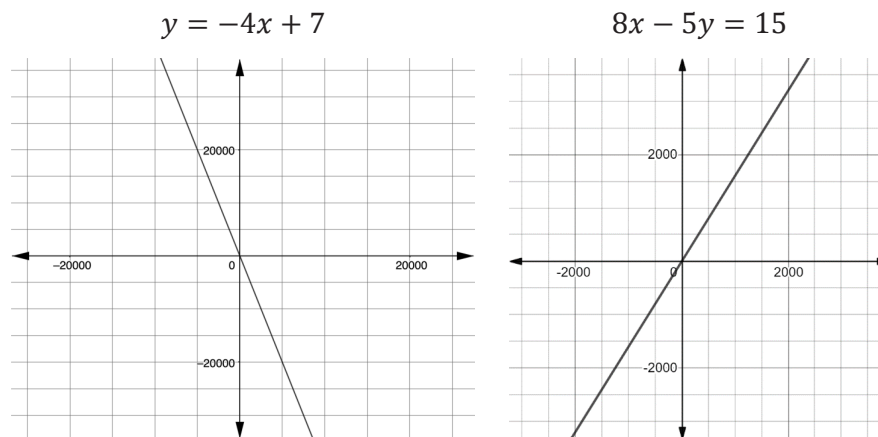
7. Based on the definition of range, for the linear relationship $8x - 5y = 15$, what values are outputs for y ?

- Ⓐ all values greater than -10 and less than 10
- Ⓑ all values greater than -20 and less than 20
- Ⓒ it cannot be determined
- Ⓓ any value



4.1.3 Guided Instruction: Set-Builder Notation

The graphs of $y = -4x + 7$ and $8x - 5y = 15$ are shown where the graph has been zoomed out.



The domain and range of a linear relationship are all **real numbers**.

By zooming out, it is clearer that the domain and range of $y = -4x + 7$ and $8x - 5y = 15$ is all real numbers.

There are several ways to mathematically represent the domain and range of all real numbers.

	Written Description	Inequalities	Set-Builder Notation
Domain	All real numbers	$-\infty < x < \infty$	$\{x: x \in \mathbb{R}\}$ or $\{x x \in \mathbb{R}\}$
Range	All real numbers	$-\infty < y < \infty$	$\{y: y \in \mathbb{R}\}$ or $\{y y \in \mathbb{R}\}$

The representation of the domain using inequalities is read as, “negative infinity is less than x , which is less than infinity.”

Notice that the inequality sign used with infinity does not include an equal sign.

1. Discuss with your partner why the domain $-\infty < x < \infty$ cannot be written as $-\infty \leq x \leq \infty$.



Set-builder notation is a shorthand used to write sets, often sets with an infinite number of elements. The set $\{x: x > 0\}$ is read aloud, “the set of all x , such that x is greater than 0.” It is read aloud exactly the same way when the colon (:) is replaced by the vertical line (|) as in $\{x|x > 0\}$.

The table shows number sets within the number system, examples of the types of numbers included in each set, and the symbol used in set-builder notation, if there is one.

	Examples	Symbol
Natural numbers	1, 2, 3, 4, ...	$\mathbb{N}$
Whole numbers	0, 1, 2, 3, ...	No symbol
Integers	..., -4, -3, -2, -1, 0, 1, 2, 3, 4, ...	$\mathbb{Z}$
Rational numbers	$-20\frac{7}{8}, -\frac{2}{7}, -0.25, \sqrt{9}, 4.01, \sqrt[3]{125}, 14\frac{5}{9}, 89$	$\mathbb{Q}$
Irrational numbers	$-\sqrt{17}, \pi, \frac{\sqrt{6}}{2}, \sqrt[3]{28}$	No symbol
Real numbers	All rational and irrational numbers	$\mathbb{R}$

In most cases, set-builder notation is similar to an inequality, except when referring to a number system or a specific pattern of numbers.

2. Work with your partner to complete the table. The first row has been completed.

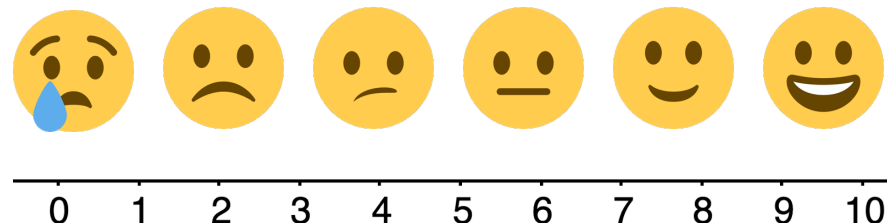
Written Description	Inequalities	Set-Builder Notation
All numbers greater than or equal to negative three	$x \geq -3$	$\{x: x \geq -3\}$ or $\{x x \geq -3\}$
All numbers less than seven		
All numbers greater than or equal to negative two and less than or equal to eleven		
All numbers greater than zero and less than or equal to twenty		
The set of integers		
The set of rational numbers		

3. Explain why writing the set of integers and the set of rational numbers using inequalities is not possible.



4.1.4 Wrap-Up: Reflection

1. Use the emoji scale to describe how well you understand the domain and range of a linear relationship.



2. Use the emoji scale to describe how well you understand set-builder notation.

